

WEEKLY REPORT 3

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Week: 2 (Super-Resolution Implementation Phase) Project Title: Super Resolution using Classical Interpolation Methods

This week had the goal of transforming the project to satellite image super-resolution when compared to generic image super-resolution making sure that the data reflects real-life remote sensing data. There was also a goal of constructing an adaptive SR pipeline, which can meet any image resolution dynamically without the constraints imposed by fixed ratios of upscale, which was the case in the previous weeks.

Work Done

a. Dataset Preparation

We have just completed the transition to the satellite image world this week. The team had visited several open-access repositories such as NASA Worldview, Copernicus Open Data Hub and USGS Earth Explorer in its quest to obtain real time high-resolution satellite images.

A selective set of 10 satellite images was established, by representing different landscapes:

Cities (streets, constructions)

Forest and vegetation areas

Agricultural zones

Coastal regions

The mountainous and desert states.

All the images were scaled and converted to .png to ensure that they were compatible with the openCV and python processing libraries. The data now constitutes the starting point of all the further SR experiments.

b. Code Implementation of adaptive SR Code.

Our super-resolution pipeline was re-designed with no scaling factors that were fixed.

Rather, the optimum downscale factor is now dynamically calculated by the system depending on the size of the image.

This logic ensures:

Small images ($\leq 500\text{px}$): downscale $\times 2-3$

Medium images ($500-2000\text{px}$): downscale $\times 4-6$

Large images ($> 2000\text{px}$): downscale $\times 8$

This dynamic process enables our algorithm to be operated in various resolutions without a human control. The interpolation, reconstruction and assessment steps are automated.

Output Generation and Verification: Staff must thoroughly understand system verification to guarantee the correct functionality of the new system. Generation and Verification of output: The staff should be well informed on system verification to ensure that the new system meets the right functionality.

Adaptive system produced LR images and SR images of all samples.

Ensured that the SR images had spatial integrity and geometry was sound.

Calculated PSNR and SSIM of each interpolation (Nearest, Bilinear, Bicubic) to verify that the baseline is correct.

Noticed the gradual increase in the two measures as a result of adaptive factor selection.

Observations

Moderate ($\times 4, \times 6$) scaling worked better than extreme ($\times 8$) on large satellite images.

Bicubic interpolation was still seen to offer the best initial quality to be further refined.

The process of adaptive scaling was found to be a huge enhancement of the uniformity of performance on various images.

Challenges

Google Colab would develop memory problems occasionally when dealing with large satellite files ($>30\text{MB}$).

The aspect ratios and non-resampling distortions necessitated a further verification.

Next Week Plan

Improve non-ML reconstruction technology:

Edge-preservation Edge-Directed Interpolation (NEDI).

Refinement in terms of an Iterative Back Projection (IBP).

Enhancement using wavelet, to sharpen a texture.

Start preliminary invasion of the gearing pipeline with a fraction of the data.

Make comparative measures based on the interpolation baseline.